Lecture 27

Exam: May 15, 2-sided cheat, cumulative 50% prob., practice exam.
10q — 3hr.

So far: S, $E \subseteq S$, $\mathbb{P}(E)$, $\mathbb{P}(E|F)$ $\vdots$ $X : S \to \mathbb{R}$

$$\mathbb{E}X = \sum_r r\,\mathbb{P}(X=r)$$

$$\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y.$$

$$V(Y) = \mathbb{E}(Y - \mathbb{E}Y)^2$$

I Markov's Ineq II Chebyshev's Ineq ~ "stdev(Y) ~ typical fluctuation"

Motivating: How do you estimate the outcome of an election without counting all the votes?

I Markov.

Q) If $X : S \to \mathbb{R}$ random variable, how close is X typically to $\mathbb{E}X$?

Theorem: If $Y : S \to \mathbb{R}_{\geq 0}$ is a nonnegative random variable, then

$\forall t > 0$

$$\boxed{\mathbb{P}(Y \geq t) \leq \frac{\mathbb{E}Y}{t}}$$

Nontrivial when $t > \mathbb{E}Y$.

"A nonnegative r.v. is unlikely to be much larger than its average."

ex1: Y = random Midterm 2 Score.

$\mathbb{E}Y = 30$.

$t = 40$:

inequality.

$$\mathbb{P}(Y \geq 40) \leq \frac{\mathbb{E}Y}{40} = \frac{30}{40} = \underline{\underline{0.75}}.$$

$\therefore$ At most 75% of the class got ≥ 40.

(Q) When is this sharp?

(A) Suppose $Y = \begin{cases} 0 & \text{with prob } 1/4 \\ 40 & 3/4. \end{cases}$

$$\mathbb{E}Y = \frac{1}{4}\cdot 0 + \frac{3}{4}\cdot 40 = \underline{\underline{30}}$$

$$\mathbb{P}(Y \geq 40) = .75.$$

<u>Proof:</u> Let $Y \geq 0$, $t > 0$.

$$\mathbb{E}Y \overset{\text{defn}}{=} \sum_{r \in \text{range}(Y)} r\,\mathbb{P}(Y = r)$$

$$= \sum_{r \geq t} r\,\mathbb{P}(Y=r) + \underbrace{\sum_{\substack{r<t \\ r\geq 0}} r\,\mathbb{P}(Y=r)}_{\geq 0}$$

$$\overset{\text{nonneg.}}{\geq} \sum_{r \geq t} r\,\mathbb{P}(Y=r)$$

$$\overset{(r \geq t)}{\geq} \sum_{r \geq t} t\,\mathbb{P}(Y=r)$$

$$= t \sum_{r \geq t} \mathbb{P}(Y=r) = t\,\mathbb{P}(Y \geq t).$$

Rearrange: $$\mathbb{P}(Y \geq t) \leq \frac{\mathbb{E}Y}{t}. \qquad \square$$

Remark: $t = \mathbb{E}Y$: $\boxed{\mathbb{P}(Y \geq \mathbb{E}Y) \leq 1.}$ trivial.

However: cf: $\exists s \in S \; (Q(s))$

$$\uparrow$$

$$\mathbb{P}(Q(s)) > 0$$

$$\uparrow$$

$$\mathbb{E} \; 1_{Q(s)} > 0.$$

Used $\boxed{\mathbb{P}(Y \geq \mathbb{E}Y) > 0 .}$

Proof: Assume $\mathbb{P}(Y \geq \mathbb{E}Y) = 0.$

$$\mathbb{E}Y = \sum_{r < \mathbb{E}Y} r \, \mathbb{P}(Y = r)$$

$$< \mathbb{E}Y \sum_r \mathbb{P}(Y=r) = \mathbb{E}Y.$$

contradiction $\square$

II Chebyshev.

Theorem: If Y is a random variable,

Variance.

$$\forall t>0 \quad \boxed{\mathbb{P}(|Y-\mathbb{E}Y| \geq t) \leq \frac{V(Y)}{t^2}}$$

2-sided.

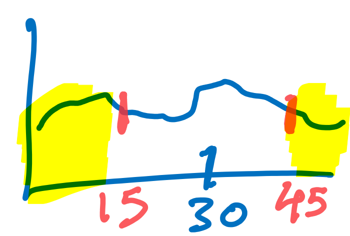
Nontrivial when $t > \sqrt{V(Y)} = \text{stdev}(Y)$.

"A random variable is unlikely to deviate from its mean by much more than its stdev."

ex2: $Y =$ random midterm 2 score in this class.

$\mathbb{E}Y = 30.$

$\text{stdev}(Y) = 10 = \sqrt{V(Y)}.$



Chebyshev:

$t = 15$

$$\mathbb{P}\left(|Y-30| \geq 15 \right) \leq \frac{100}{15^2} = \frac{100}{225} < 50\%.$$

At most $\frac{100}{225}$ of the class deviated by more than 15.

Proof: Given Y, $t > 0$, let $\mu_Y = \mathbb{E}Y$.

Define $Z = (Y-\mu_Y)^2$, nonnegative.

$$\mathbb{E}Z = \mathbb{E}(Y-\mu_Y)^2 = V(Y).$$

$$\mathbb{P}(|Y-\mu_Y| \geq t) \overset{\text{(nonneg)}}{=} \mathbb{P}\left((Y-\mu_Y)^2 \geq t^2\right)$$

$$= \mathbb{P}(Z \geq t^2)$$

$$\overset{\text{Markov}}{\leq} \frac{\mathbb{E}Z}{t^2} = \frac{V(Y)}{t^2}. \quad \square$$

$a, b \geq 0$

$a \geq b \iff a^2 \geq b^2$

Simplest Stat Problem:

Given a coin with $\boxed{\text{unknown}}$ bias $q \in [0,1]$, how do you find q?

idea: Flip the coin n times independently.

Let $X_i = \begin{cases} 1 & \text{if } i\text{th flip } H \\ 0 & \text{" } T \end{cases}$, $i = 1, \dots, n$

independent.

Define $\hat{q} = \dfrac{X_1 + \cdots + X_n}{n}$

fraction of flips that are H

random variable. "estimator"

Observe: 1) $\mathbb{E}\hat{q} = \dfrac{\mathbb{E}X_1 + \cdots + \mathbb{E}X_n}{n} = \dfrac{q + \cdots + q}{n} = q$.

② $V(\hat{q}) \overset{\text{defn}}{=} \mathbb{E}\left(\frac{X_1+\cdots X_n}{n} - \frac{nq}{n}\right)^2$

$$= \frac{1}{n^2}\,\mathbb{E}\left((X_1+\cdots X_n) - nq\right)^2$$

$$= \frac{1}{n^2}\,V(Y) \qquad Y \sim \text{Binomial}(n,q)$$

$$\overset{\text{lec26}}{=} \frac{1}{n^2}\cdot nq(1-q) = \frac{q(1-q)}{n}.$$

<u>Observe:</u> $\frac{q(1-q)}{n} \leq \frac{1}{4n}$ for all $q \in [0,1]$.

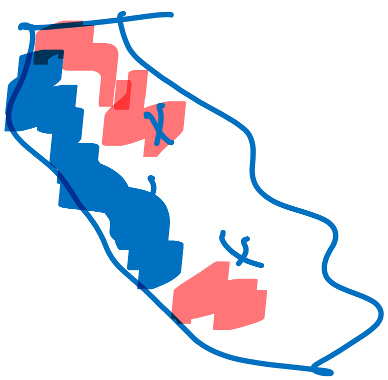
<u>Chebyshev:</u> $P(|\hat{q}-q| \geq t) \leq \frac{V(\hat{q})}{t^2} \leq \frac{1}{4nt^2}.$

plug: $t = \frac{1}{20}$, $n = 1000$: $\mathbb{P}\left(|\hat{q} - q| \geq \frac{1}{20}\right) \leq \frac{1}{4 \cdot 1000 \cdot \frac{1}{400}}$

$$= \frac{1}{10}.$$

$\therefore$ With 90% probability, $|\hat{q} - q| < \frac{1}{20}$.

Election: 2024 Election. 40 Million voters CA.



Say q fraction voted for trump.

How to find q? Count all votes $\rightsquigarrow$ slow.

Experiment: Choose n voters u.a.r. independently.

Let $X_i = \begin{cases} 1 & \text{if voter } i \text{ T} \\ 0 & \text{'' B} \end{cases}$ $i = 1 \ldots n$.

$\mathbb{P}(X_i = 1) = q$.

This is the same as the coin problem.

$$\hat{q} = \frac{X_1 + \cdots + X_n}{n}.$$

By previous, $\mathbb{P}\left(|\hat{q} - q| \geq \frac{1}{20}\right) \leq \frac{1}{10}$ for $n = 1000$.

Can estimate q within 5% using 1000 voters, 90% prob.
↳ independent of population

Caveat: • Hard to draw a uniform sample.
• Margin in CA ~20% ⟶ 5% error ok.
Margin in PA ~2% ⟶ $n = 10{,}000$?

(Q) Does this hold for 3 outcomes?

(A) $Y = \begin{cases} 1 & q_1 \\ 2 & q_2 \\ 3 & q_3. \end{cases}$ $\underline{\underline{\text{want}}}$ not enough to estimate $\mathbb{E}Y$.

$X_1 = 1\{Y=1\}$ $\quad X_2 = 1\{Y=2\}$

$X_3 = 1\{Y=3\}$.

(Q) 5 candidates, some candidate has 0.01% votes.

(A) Detect & throw away.

AI.

How do you generate text?

This class is ZS ________ .

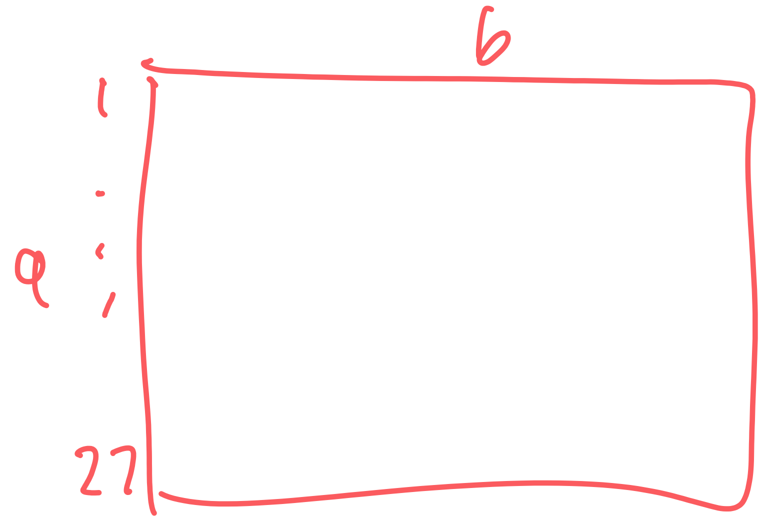
Shannon: "Shannon's Ladder"

level 0: just generate random characters from alphabet.

level 1: generate letters with prob $\propto$ freq. in a book.

level 2: Pick a random sequence of 2 letters from the book: $L_1 L_2$.

Make a table of $\boxed{\mathbb{P}(L_2 = b \mid L_1 = a)}$



27^2.

level 3

pick 3 random consecutive $L_1 L_2 L_3$.

Make a table: $P(L_3 = c \mid L_1 = a \wedge L_2 = b)$.

27^3

Words level 1: 50k

Words level 2: $(50,000)^2$.

1lan 10^6.

$(50{,}000)^{10^6} \ggg$ universe.

Cannot store table.

Replace by a function — NN.